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expressions

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A recycling bin made by a UAE University team responds with a smile as soon as someone drops a bottle

> We found out that compared with the normal recycling bin, recycling rate went up three times with the smiling face
> **Dr Jose Berengueres**

Wadeema Fahad AlKaabe, a student of Al Raqiah School, watches a smiley face on the screen as her schoolmate Raouda Abdullah Almazrai throws a bottle inside the box at their school in Al Ain. Right, Jose Berengueres, Assistant Professor, talking about the smiling bin. — *KT photos by Nezar Balout*

Olivia Olarte-Ulherr

What if you receive a smile every time you recycle? Is that enough motivation for you to go green?

A coin-tinkling three times YES, according to findings of an experiment conducted at schools in Al Ain.

For this project, a team from the UAE University (UAEU) made use of an Android tablet, proximity sensor and speaker, and attached this to a standard recycling bin.

Whenever someone drops a polythene terephthalate (PET) bottle in the bin, the sensor is triggered and this produces the Nintendo coin sound and the poker face on the screen changes into the Japanese smiley emoticon, rewarding user with a happy face.

"The simple smile is the reward... As human, we respond very much to a smile," stated Dr Jose Berengueres, Assistant IT Professor at UAEU who headed the creative team. And to prove this, the team placed two recycling bins — an emoticon bin and standard recycling bin — next to each other at the College of IT lobby.

"We found out that compared with the normal recycling bin, recycling rate went up three times with the smiling face," he said, which proved that users prefer to be rewarded with a smile.

This simple yet innovative creation also proved that the 'feel good' factor of a smile is a novel way to effect good practice and foster positive behaviour among the younger students.

An emoticon bin was placed in February at the canteen of the Al Raqia School for girls.

This became instantly popular with the Cycle 1 (Grades 1-5) students that in their excitement, they brought boxes of bottles from home the next day, "which was not the point," related the amused Eeva-Maria Eravalo, a teacher at Al Raqia school, who organised the placement of the smiling bin at her school. After one week, however, the emoticon bin failed to work and the kids refused to recycle because it doesn't smile.

"When the smiling face was not working, the girls did not bother to go all the way to the canteen to put the bottles in the smiling bin," Eravalo explained.

According to her, the school already has a culture of recycling, but the emoticon bin motivated the girls to recycle more "because it's different."

"Within the first week we gathered about 120 bottles and about 300 in a month," Eravalo said.

Dr Berengueres said the idea for his brain child came after a sustainability report in 2010 revealed that out of the 750 million water bottles consumed every year (third highest in the world) in the UAE, only about 10-12 per cent are being recycled.

"This means for every 10 bottles, only one will be recycled. The survery also showed that only 12 per cent of parents tell their kids to recycle," Dr Berengueres, who is also the director of the UAEU robotics lab, said.

This is compared to other countries such as Japan, which recycles 77 per cent of the PET bottles and the US at 27 per cent.

The first prototype took about three months and initially used a PC and LCD screen.

Emirati student Fatma Al Suwairi wrote the software code that produced the smiley face response and other students helped to analyse the research findings.

Tony Ng, an instructor at UAEU, made the Android version of the software, which connects wirelessly to the sensor.

With the use of Android tablet, the cost of the entire project went down to about $150 (Dh552) — $100 for the tablet and $50 for the Lego mindstorm sensor. And since the app was made available by the team as free download, the smiling bin is affordable and a more effective way to advertise and raise awareness about the importance of recycling.

However, the question is, if you are consuming electricity 24 hours a day to power up the tablet, isn't that opposed to being green?

"We did a calculation," said Dr Berengueres.

According to their calculation, based on a 500ml PET bottle at €1.5 (Dh7), three bottles are needed to offset a day of electric bill in the UAE, seven bottles in the US while 19 bottles in Germany.

"You can recover your investment in one year. There's a global market for recycling plastic," Dr Berengueres pointed out.

During the Human Robot Interaction conference in Tokyo, Japan last month, the UAEU's smiling bin was awarded second prize for best video. Several entities have already expressed interest in the team's creation.

A simple smile can truly go a long way, and it does not have to go just in your direction.

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